

*This question paper contains 4 printed pages.*

*Your Roll No. ....*

S. No. of Paper : 6633 HC  
Unique paper code : 32351402  
Name of the paper : Reimann Integration and Series of Functions  
Name of course : B.Sc. (Hons.) Mathematics  
Semester : IV  
Duration : 3 hours  
Maximum marks : 75

*(Write your Roll No. on the top immediately on receipt of this question paper.)*

*Attempt any two parts from each question.  
All questions are compulsory.*

Q.1 (a) Prove that a bounded function  $f$  on  $[a, b]$  is integrable if and only if for each  $\varepsilon > 0$ ,  $\exists$  a partition  $P$  of  $[a, b]$  such that  $U(f, P) - L(f, P) < \varepsilon$ . (6)

(b) Show that if  $f$  is integrable on  $[a, b]$ , then  $|f|$  is integrable on  $[a, b]$ , and:

$$\left| \int_a^b f \right| \leq \int_a^b |f|.$$

Hence, show that :

$$\left| \int_{-2\pi}^{2\pi} x^2 \sin^8(e^x) dx \right| \leq \frac{16\pi^3}{3}. \quad (6)$$

(c) Suppose  $f$  and  $g$  are continuous functions on  $[a, b]$ , and  $g(x) \geq 0 \quad \forall x \in [a, b]$ . Prove that,  $\exists x \in [a, b]$  such that:

$$\int_a^b f(t)g(t) dt = f(x) \int_a^b g(t) dt. \quad (6)$$

Q.2 (a) (i) The Dirichlet function  $f: [0, 1] \rightarrow \mathbf{R}$  is defined by :

$$f(x) = \begin{cases} 1, & x \in [0, 1] \cap \mathbf{Q} \\ 0 & x \in [0, 1] \setminus \mathbf{Q} \end{cases}$$

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Is  $f$  Riemann Integrable? Justify.

(ii) Show that a decreasing function on  $[a, b]$  is integrable.

(b) (i) State Fundamental Theorem of Calculus-I.

(ii) Let  $f$  be an integrable function on  $[a, b]$ . Let  $P$  be a partition of  $[a, b]$  and  $P^*$  be a refinement of  $P$  such that  $P^* = P \cup \{u\}$ . Show that  $L(f, P) \leq L(f, P^*)$ .

(c) (i) For a bounded function  $f$  on  $[a, b]$ , define the Riemann Sum associated with a partition  $P$ . Hence, give Riemann's definition of integrability.

(ii) Calculate  $\lim_{h \rightarrow 0} \frac{1}{h} \int_3^{3+h} e^{t^2} dt$

3. (a) Show that:

$$\beta(x, y) = \int_0^1 t^x (1-t)^y dt \text{ converges } \Leftrightarrow x > 0, y > 0.$$

(b) (i) Define Improper Integral of Type-II. When does it converge? When is it said to diverge?

(ii) Determine if  $\int_{-\infty}^{\infty} e^x dx$  is a convergent integral. Justify.

(c) (i) Determine if the following integrals are Improper Integrals. If so, what kind? Justify.

$$\int_0^{\pi/2} \frac{\sin x}{\sqrt[3]{x}} dx, \quad \int_0^1 x \ln(x) dx$$

(ii) Prove that  $\int_{\pi}^{\infty} \frac{\sin x}{x} dx$  converges absolutely.

4. (a) State and prove Cauchy's criterion for uniform convergence for sequences of functions.

(b) (i) Is the sequence  $\{f_n\}$  where :

$$f_n = \frac{1}{n} \sin(nx + n).$$

uniformly convergent on  $\mathbf{R}$ ? Justify. (2)

(ii) Suppose a sequence  $\{f_n\}$  converges uniformly to  $f$  on a set  $A$ , and further suppose that each  $f_n$  is bounded on  $A$ . Show that the limit function  $f$  is bounded on  $A$ . (4)

(c) Show that if  $a > 0$ , then the sequence  $\{n^2 x^2 e^{-nx}\}$  converges uniformly on the interval  $[a, \infty)$ , but that it does not converge uniformly on the interval  $[0, \infty)$ . (6)

5. (a) If  $f_n$  is continuous on  $D \subseteq \mathbf{R}$ , for each  $n \in \mathbf{N}$  and if  $\sum f_n$  converges to  $f$  uniformly on  $D$ , then  $f$  is continuous on  $D$ . (6)

(b) Show that the series of functions  $\sum \frac{x^n}{(1+x^n)}$ ,  $x \geq 0$ , converges uniformly on  $[0, a]$  for  $0 < a < 1$ , but is not uniformly convergent on  $[0, 1)$ . (6)

(c) A sequence  $\{f_n\}$  converges uniformly to  $f$  on  $A_0$ , if for each  $\varepsilon > 0$  there is a natural number  $K(\varepsilon)$ , depending only on  $\varepsilon$ , such that if  $n \geq K(\varepsilon)$ , then:

$$|f_n(x) - f(x)| < \varepsilon \quad \text{for all } x \in A_0.$$

Hence state a necessary and sufficient condition for a sequence  $\{f_n\}$  to fail to converge uniformly on  $A_0$  to  $f$ .

Apply this condition on sequence  $f_n(x) = \frac{x}{n}$ , for  $x \in \mathbf{R}$  to examine for uniform convergence. (6)



6. (a) (i) Suppose that  $f(x) = \sum_{n=0}^{\infty} a_n x^n$  has radius of convergence  $R > 0$ . Then show that  $\int_0^x f(t) dt = \sum_{n=0}^{\infty} \frac{a_n}{n+1} x^{n+1}$  for  $|x| < R$

(5)

(ii) State when is a power series differentiable term by term. (1.5)

(b) (i) Apply Abel's Theorem to show :

$$\log_e 2 = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} \dots \quad (3.5)$$

(ii) Given  $\frac{1}{1-x} = \sum_{n=0}^{\infty} x^n$  for  $|x| < 1$ . Evaluate:

$$\sum_{n=1}^{\infty} \frac{n}{5^n}, \quad \sum_{n=1}^{\infty} \frac{n^2 (-1)^n}{3^n}. \quad (3)$$

(c) (i) Apply Ratio Test for series to show that radius of convergence  $R$  of the power series  $\sum a_n x^n$  is given by

$$\lim_{n \rightarrow \infty} \left| \frac{a_n}{a_{n+1}} \right|, \text{ whenever the limit exists.} \quad (3.5)$$

(ii) Find radius of convergence for :

$$\sum_{n=2}^{\infty} (\ln(n))^{-1} x^n, \quad \sum_{n=0}^{\infty} \frac{(n!)^3 x^n}{(3n)!} \quad (3)$$